

Strategy Pickers: Addition Facts

Answer Key

Kindergarten–Grade 1

Quick Answers

- | | | |
|------|-------|--------------------|
| 1. 9 | 5. 10 | 9. 8 |
| 2. 8 | 6. 10 | 10. the sum = 9, — |
| 3. 9 | 7. 10 | |
| 4. 8 | 8. 7 | |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

1.OA.C.6 – *problems 1, 2, 3, 4, 6, 8, 9, 10*

K.OA.A – *problems 5, 7*

Difficulty 1–4 of 5 across 11 tagged checks.

- 1.** Warm-up. Just one more! Count on. $8 + 1$

Solution: Adding 1 is one hop: say 8, hop 9. So $8 + 1 = 9$.

9

- 2.** Warm-up. Count on from 6. $6 + 2$

Solution: Say 6, hop 7, 8 — two hops for the 2. So $6 + 2 = 8$.

8

- 3.** Warm-up. Start with the bigger number, then count on. $2 + 7$

Solution: Addition works in either order, so start big: say 7, hop 8, 9 — only two hops instead of seven. So $2 + 7 = 9$.

9

- 4.** Pick your strategy, then add. $4 + 4$

Solution: The numbers match — it is a double. Doubles are facts to know by heart: $4 + 4 = 8$.

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- 5.** Pick your strategy, then add. $3 + 7$

Solution: 3 and 7 are tens partners — they fit together to fill a ten-frame. So $3 + 7 = 10$.

10

- 6.** Pick your strategy, then add. $5 + 5$

Solution: A double that is also a tens partner pair: $5 + 5 = 10$.

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7. Pick your strategy, then add. $6 + 4$

Solution: 6 and 4 are tens partners: 6 needs exactly 4 to fill up to ten. So $6 + 4 = 10$.

10

8. Pick your strategy, then add. Hint: it is almost a double! $3 + 4$

Solution: Near double: $3 + 4$ is the double $3 + 3$ and 1 more. $3 + 3 = 6$, and one more is 7. So $3 + 4 = 7$.

7

9. Pick your strategy, then add. $5 + 3$

Solution: The 3 is small, so count on from 5: say 5, hop 6, 7, 8. So $5 + 3 = 8$.

8

10. Challenge! Add $4 + 5$ with the strategy you like best. Then tell which strategy you picked and why it is quick for these numbers.

Solution: One quick way is the near double: $4 + 5$ is $4 + 4$ and 1 more, so $8 + 1 = 9$. Counting on from 5 also works: 6, 7, 8, 9. Either way, $4 + 5 = 9$.

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Tell which strategy and why (instructor judges): any named strategy that fits the numbers counts — for example “near double, because 4 and 5 are neighbors, so I did $4 + 4$ and one more” — as long as the reason matches what the student actually did.